

Algebra/Topology Seminar

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A POSITIVE FORMULA INVOLVING PIPES AND BRUHAT ORDER

Thursday, September 10, 2026
3:00 p.m. in Massry B008

ABSTRACT. Positivity phenomena are common in algebraic geometry: coefficients coming from complicated expansions often turn out to be nonnegative integers, suggesting that they should count something. In this talk, I will discuss one such problem arising from Schubert calculus. Lam, Lee, and Shimozono proved certain coefficients in their study of affine Grassmannians are positive and asked for a combinatorial explanation. I will present the first such formula, using two combinatorial models: bumpless pipe dreams, which are configurations of pipes on a grid, and the Bruhat order. I will also explain how a correspondence between these models leads to the positivity formula. This is joint work with Jack Chou.